

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, v_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, v_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$$

Span

$$\begin{aligned} & \text{Span}(V_1, V_2, V_3) \\ &= c_1 V_1 + c_2 V_2 + c_3 V_3 \\ &= \underline{c_1} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \underline{c_2} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} + \underline{c_3} \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} \end{aligned}$$

linear comb.

Spanning set

$$\text{span} = \left\{ v_1, v_2, v_3 \right\}$$

$$= \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 11 \\ 8 \\ 9 \end{bmatrix} \right\}$$

ex.

(v ₁) drink x1	+ v2
(v ₂) rice x2	+ x1
(v ₃) apple x3	+ x1

" "

Q Q

4k subSpace = $\left. \begin{array}{l} \text{low space} \\ \text{col space} \\ \text{kernel null space} \\ \text{kernel} \end{array} \right\}$

Row space

$$RS(A) = \{ \text{span of row vectors} \}$$

$$\begin{aligned} &= \text{span}(A_{(1)} \dots A_{(n)}) \\ &= c_1 A_{(1)} + \dots + c_n A_{(n)} \\ &= [c_1 \dots c_n] \begin{bmatrix} A_{(1)} \\ \vdots \\ A_{(n)} \end{bmatrix} \end{aligned}$$

$$\text{Span} = \{xA \mid x \in \mathbb{R}^{1 \times n}\}$$

linear comb.

Column space

$$CS(A) = \{ \text{span of column vectors} \}$$

$$= \text{span}(A^{(1)} \dots A^{(m)})$$

$$= [A^{(1)} \dots A^{(m)}] \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix}$$

$$= \{Ax \mid x \in \mathbb{R}^{n \times 1}\}$$

$\longleftrightarrow$
linear
comb.

$$\underline{A} \in \mathbb{R}^{m \times n}$$

(1) Seen as matrix

$$A = \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{array} \right]$$

(2) seen as set of low space $\left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right]$

$$A = \begin{bmatrix} A^{(1)} \\ A^{(2)} \\ A^{(3)} \end{bmatrix}, \text{ where } A^{(1)} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

3) seen as set of column space $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$A = [A_{(1)}, A_{(2)}, A_{(3)}]$$

Spanning set

where $A_{(1)} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $A_{(2)} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$, $A_{(3)} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$

(4) seen as block matrix $\begin{bmatrix} B & C \\ D & E \end{bmatrix}$ # recursive

$$A = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$$

where $X = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, $Y = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

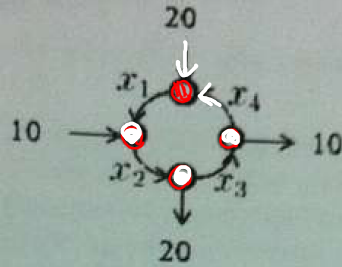
$$Z = [0 \ 0], W = [3]$$

ch1: system of linear eq.s

ch2: det

ch3: vector

$\sum \vec{in} = \sum \vec{out}$



$$\begin{cases} 20 + x_4 = x_1 \\ 10 + x_1 = x_2 \\ x_2 = 20 + x_3 \\ x_3 = 10 + x_4 \end{cases}$$

var. const.

$$\begin{cases} x_1 - x_4 = 20 \\ x_1 - x_2 = -10 \\ x_2 - x_3 = 20 \\ x_3 - x_4 = 10 \end{cases}$$

$x_2 - x_4 = 30$

$$\begin{array}{cccc|c} x_1 & x_2 & x_3 & x_4 & \\ \hline 1 & 0 & 0 & -1 & 20 \\ 1 & -1 & 0 & 0 & -10 \\ 0 & 1 & -1 & 0 & 20 \\ 0 & 0 & 1 & -1 & 10 \end{array}$$

Sol

elementary row op.

- (i) swap (row- $\vec{i}$, row- $\vec{j}$) , $r_{\vec{i}\vec{j}}$
- (ii) $k \cdot \text{row-}\vec{i}$, $r_{\vec{i}}^{(k)}$
- (iii) $\text{row-}\vec{j} = \text{row-}\vec{j} + k \cdot \text{row-}\vec{i}$, $r_{\vec{i}\vec{j}}^{(k)}$

Gauss-jordan elimination (RREF)

(-1)

$$\begin{array}{cccc|c} x_1 & x_2 & x_3 & x_4 & \\ \hline 1 & 0 & 0 & -1 & 20 \\ 1 & -1 & 0 & 0 & -10 \\ 0 & 1 & -1 & 0 & 20 \\ 0 & 0 & 1 & -1 & 10 \end{array}$$

$$\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 20 \\ 0 & 1 & 0 & 0 & -10 \\ 0 & 0 & 1 & 0 & 20 \\ 0 & 0 & 0 & 1 & 10 \end{array}$$

(Row Echelon Form) REF

$$\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 20 \\ 0 & -1 & 0 & 1 & -30 \\ 0 & 1 & -1 & 0 & 20 \\ 0 & 0 & 0 & -1 & 10 \end{array} \rightarrow \begin{array}{cccc|c} 1 & 0 & 0 & -1 & 20 \\ 0 & -1 & 0 & 1 & -30 \\ 0 & 0 & -1 & 1 & -10 \\ 0 & 0 & 0 & -1 & 10 \end{array}$$

(i) first entry = 1

$$\begin{array}{c} \checkmark \\ \times \end{array} \begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \rightarrow \begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}$$

back substitution

(i) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\leftarrow$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\begin{matrix} \text{SI 标准阵} \\ \text{秩} = 0 \\ = 1 \\ = 3 \end{matrix}$

(ii) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\begin{matrix} \text{秩} = 0 \\ = 1 \\ = 3 \end{matrix}$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\rightarrow$ $\begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ 1 & 0 & 0 & -1 & 20 \\ 0 & 1 & 0 & -1 & 30 \\ 0 & 0 & 1 & -1 & 10 \\ 0 & 0 & 0 & 1 & -10 \end{bmatrix}$ Sack sub.

$$\begin{cases} x_4 = -10 \\ x_3 - x_4 = 10 \\ x_3 = x_4 + 10 = -10 + 10 = 0 \\ x_2 - x_4 = 30 \\ x_2 = 30 + x_4 = 30 - 10 = 20 \\ x_1 - x_4 = 20 \\ x_1 = 20 + x_4 = 20 - 10 = 10 \end{cases}$$

$$\begin{cases} x_1 = 10 \\ x_2 = 20 \\ x_3 = 0 \\ x_4 = -10 \end{cases}$$

Determinant

non singular (非奇异的)

$Ax=0$, $x=0$

singular

homo. equation

$Ax=0$, $x \neq 0$

fast method?

$\det(A) \neq 0 \Leftrightarrow \text{non singular}$

ex. $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $Ax=0$, $\text{秩} \neq x=0$ 吗?

sol) $\det \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = 1 \times 4 - 2 \times 3 = 4 - 6 = -2 \neq 0$
 $\therefore$ non singular

(-3) $\begin{bmatrix} 1 & 2 & 0 \\ 3 & 4 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & -2 & 0 \end{bmatrix}$
 $-2x_2 = 0, x_2 = 0$
 $x_1 + 2x_2 = 0, x_1 = 0$
 $\therefore x_1 = x_2 = 0$

$$1 \times 1: [2] \rightarrow \det([2]) = 2$$

$$2 \times 2: \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \times 1 - 0 \times 0 = 1$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - cb$$

$$3 \times 3: \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow (adi + bfg + chd) - (cdg + deb + afh)$$

⋮

$n \times n$: recursive det: Laplace expansion

$$\left\{ \begin{aligned} \det(A) &= \sum_{j=1}^n a_{ij} \cdot \text{cof}(a_{ij}) \\ \text{cof}(a_{ij}) &= (-1)^{i+j} \det(M_{ij}) \quad \text{eg row } i \text{ col } j \\ M_{ij} &= \begin{matrix} \text{take out row } i \\ \text{take out col } j \end{matrix} \rightarrow (n-1) \times (n-1) \text{ matrix} \\ &\text{minor} \end{aligned} \right.$$

ex,

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\begin{aligned} \det(A) &= a_{33} \times \text{cof}(a_{33}) \\ &= a_{33} \times [(-1)^{3+3} \det(M_{33})] \\ &= 3 \times 1 \times \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= 3 \times 1 \times (1 \times 1 - 0 \times 0) \\ &= 3 \times 1 \times 1 = 3 \end{aligned}$$

det. properties

$A^{n \times n} \leftarrow \text{square}$

row op. $\rightarrow$ low op. multiplies
 $I_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & \ddots & & \\ 0 & & \ddots & \\ 0 & & & 1 \end{bmatrix} \xrightarrow{\text{only 1 low operation}} \begin{bmatrix} 0 & \dots & 1 \\ 0 & \dots & 0 \\ \vdots & & \vdots \\ 1 & 0 & \dots & 0 \end{bmatrix} \xrightarrow{R_{1n}}$

(1) row op.

(i) $\text{swap}(\text{row}-i, \text{row}, j) \rightarrow \begin{matrix} i \rightarrow \\ j \rightarrow \end{matrix} \begin{bmatrix} 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 1 \end{bmatrix} \rightarrow \det(R_{ij} \cdot A) = \underline{(-1) \det(A)}$

(ii) $\text{row}-i \times k \rightarrow \underline{k \cdot \det(A)}$

(iii) $\text{row}-j = \text{row}-j + k \cdot \text{row}-i \rightarrow \underline{1 \cdot \det(A)}$ 不改變 $\det$.

$\star$ (2) $\det(\underline{AB}) = \det(A) \det(B)$

$\equiv$ let B do low op. (A)

不 $\rightarrow$ 補 !

(3) $A=B, \det(AA) = \det(A) \cdot \det(A) = \det(A^2)$

$\downarrow$ expand

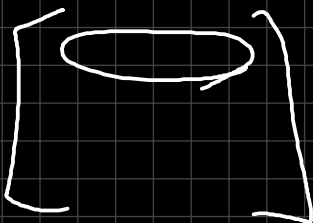
$$\det(A^k) = \det(A) \cdot \det(A^{k-1})$$
$$= \det(A) [\det(A) \cdot \det(A^{k-2})]$$

$$\vdots$$
$$= \underbrace{\det(A) \dots \det(A)}_k$$

$$= (\det(A))^k$$

$$\Rightarrow \boxed{\det(A^k) = (\det(A))^k}$$

(4) $\det(\underline{A^T}) = \det(\underline{A})$



any col.
low.

ex.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 3 \end{bmatrix}$$

det. same.

(5) w/ op.

$$\begin{array}{l} \left[\begin{array}{c} \vdots \\ 0 \end{array} \dots \begin{array}{c} \vdots \\ 0 \end{array} \right] \begin{array}{l} \text{(i)} \quad C_{ij} \\ \text{(ii)} \quad k \cdot C_{ij} \\ \text{(iii)} \quad C_j \leftarrow k \cdot C_j \end{array} \begin{array}{l} \rightarrow (-1) \det(A) \\ \rightarrow k \det(A) \\ \rightarrow 1 \cdot \det(A) \end{array} \end{array}$$

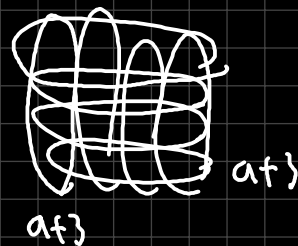
$$\left[\begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right] \xrightarrow{\text{transp}} \left[\begin{array}{ccc} \vdots & \vdots & \vdots \end{array} \right]$$

(b) diagonal matrix $D = \begin{bmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{bmatrix}$

$$\det(D) = d_1 \dots d_n$$

ex.

$$B_1 = \begin{bmatrix} a & 1 & 1 & 1 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & a \end{bmatrix} = (a+3)(a-1)^3$$



so()

* low operation (III not 1/2)

$$\begin{vmatrix} a+3 & a+3 & a+3 & a+3 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & a \end{vmatrix} = (a+3) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & a \end{vmatrix} \quad (-1)$$

$$= (a+3) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & a-1 & 0 & 0 \\ 0 & 0 & a-1 & 0 \\ 0 & 0 & 0 & a-1 \end{vmatrix} \quad M_{11}$$

$$= (a+3) \times 1 \times \begin{vmatrix} a-1 & 0 & 0 \\ 0 & a-1 & 0 \\ 0 & 0 & a-1 \end{vmatrix}$$

$$\det = (a-1) \cdots (a-1) = (a-1)^3$$

$$= (a+3)(a-1) \begin{vmatrix} a-1 & 0 \\ 0 & a-1 \end{vmatrix}$$

$$= (a+3)(a-1)(a-1)(a-1) = (a+3)(a-1)^3 \quad \#$$